Machine-Level Programming V: Advanced Topics

15-213: Introduction to Computer Systems
9th Lecture, September 26, 2017

Instructor:

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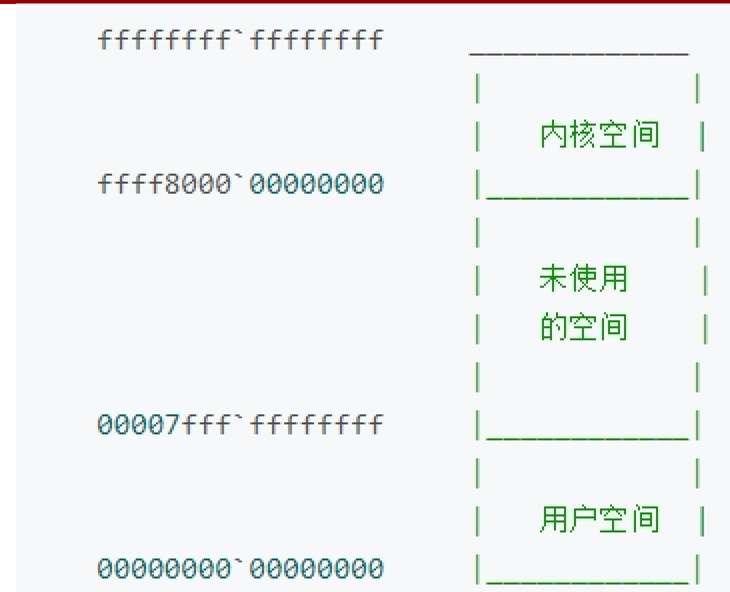
Today

- **Memory Layout**
- **Buffer Overflow**
  - Vulnerability
  - Protection
- **Unions**

X86-64linux

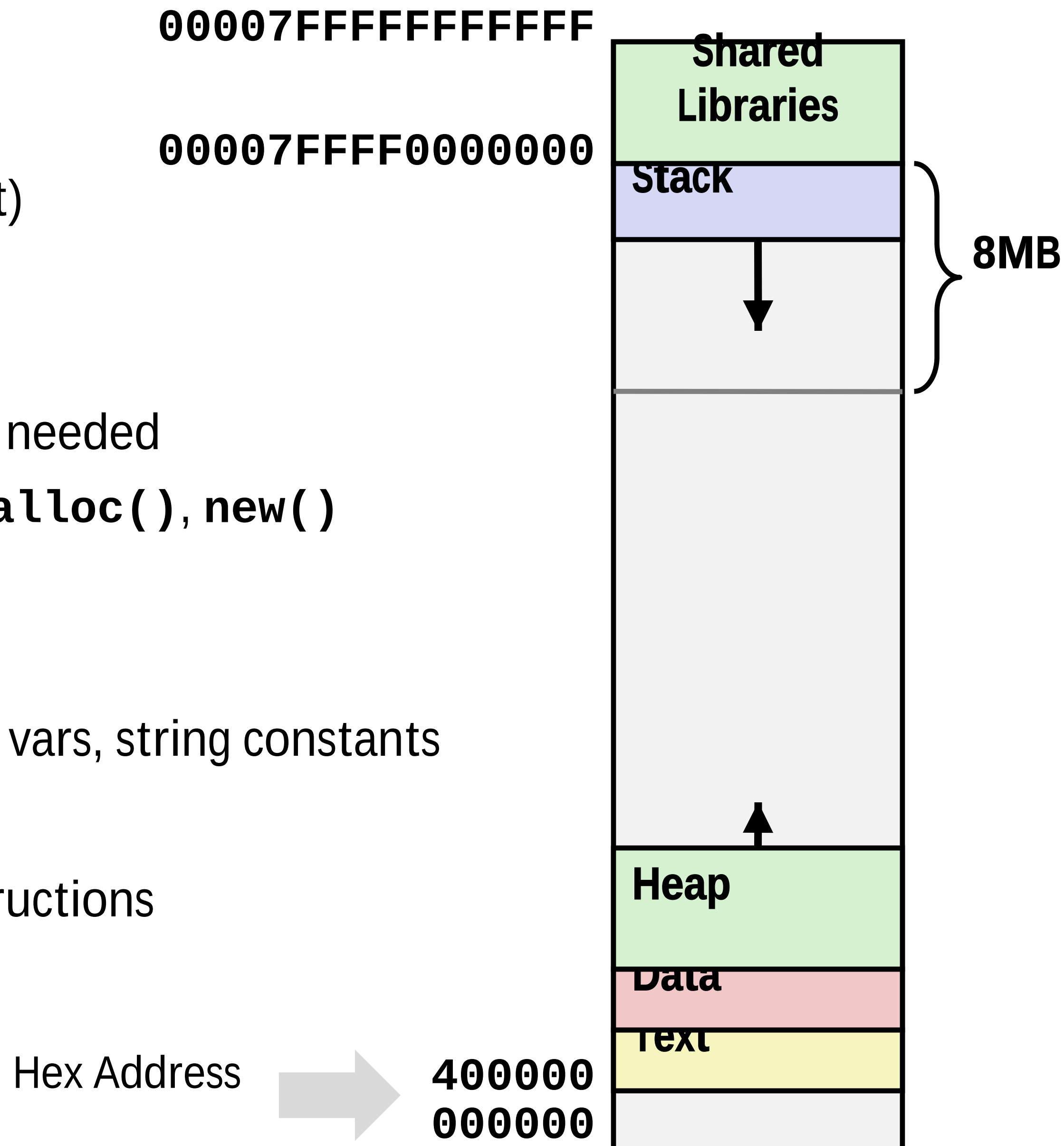


- **48位：256T**
  - 128T
  - 128T
-
  -
    - ,
    - ( )的存在,
      ,
  -
    -
  - **堆式**
    - 数据 / 无关
    - 分

x86-64 Linux Memory Layout

not drawn to scale

- **Stack**
  - Runtime stack (8MB limit)
  - E. g., local variables
- **Heap**
  - Dynamically allocated as needed
  - When call `malloc()`, `calloc()`, `new()`
- **Data**
  - Statically allocated data
  - E.g., global vars, `static` vars, string constants
- **Text / Shared Libraries**
  - Executable machine instructions
  - Read-only



Memory Allocation Example

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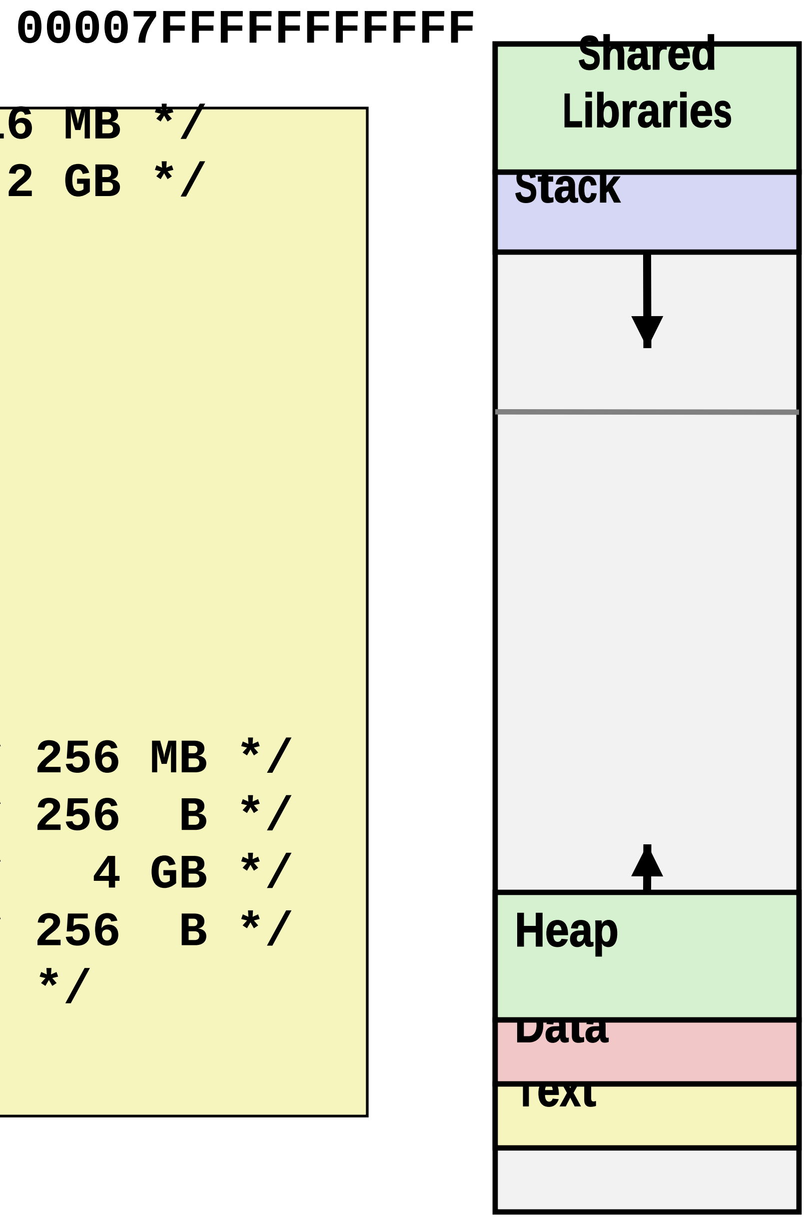
```
char big_array[1L<<24];  /* 16 MB */
char huge_array[1L<<31]; /*  2 GB */

int global = 0;

int useless() { return 0; }

int main ()
{
    void *p1, *p2, *p3, *p4;
    int local = 0;
    p1 = malloc(1L << 28); /* 256 MB */
    p2 = malloc(1L << 8);  /* 256  B */
    p3 = malloc(1L << 32); /*   4 GB */
    p4 = malloc(1L << 8);  /* 256  B */
 /* Some print statements ... */
}
```



Where does everything go?

x86-64 Example Addresses

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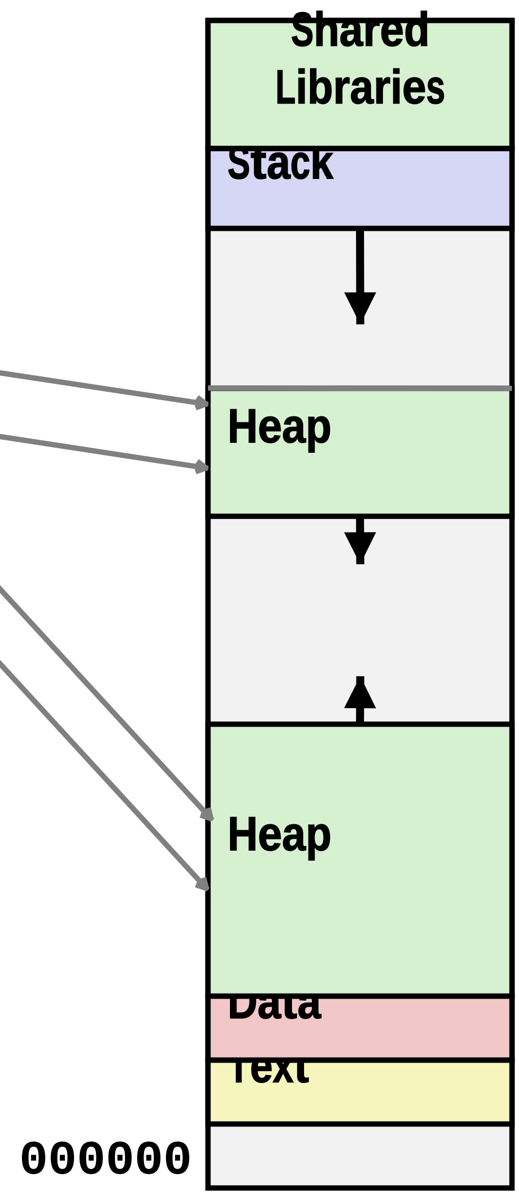
address range ~2^{47} = 128 TB

```
local  0x00007ffe4d3be87c
p1     0x00007f7262a1e010
p3     0x00007f7162a1d010
p4     0x000000008359d120
p2     0x000000008359d010
big_array     0x0000000080601060
huge_array    0x0000000000601060
main() 0x000000000040060c
useless()     0x0000000000400590
```



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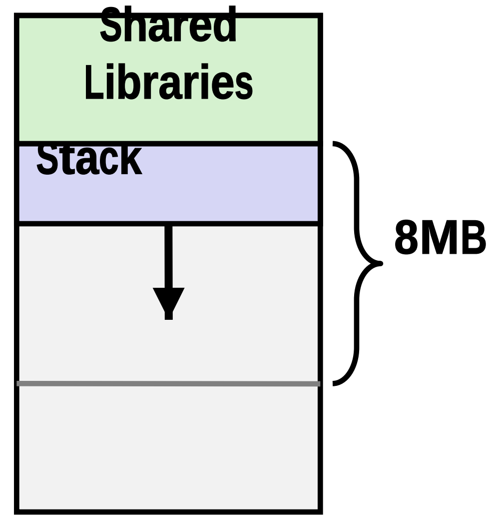
Runaway Stack Example

00007FFFFFFFFFFF



Why?

```
int recurse(int x) {
    int a[2<<15];  /* 2~17 =  128 KiB */
    printf("x = %d.  a at %p\n", x, a);
    a[0] = (2<<13)-1;
    a[a[0]] = x-1;
    if (a[a[0]] == 0)
        return -1;
    return recurse(a[a[0]]) - 1;
}
```

- Functions store local data on in stack frame
- Recursive functions cause deep nesting of frames

Why?

```
./runaway 48
x = 48.  a at 0x7fffd43e45d0
x = 47.  a at 0x7fffd43a45c0
x = 46.  a at 0x7fffd43645b0
x = 45.  a at 0x7fffd43245a0
. . .
x = 4.  a at 0x7fffd38e4310
x = 3.  a at 0x7fffd38a4300
x = 2.  a at 0x7fffd38642f0
Segmentation fault
```

Today

- Memory Layout
- **Buffer Overflow**
  - Vulnerability
  - Protection
- Unions

- CPU CPU watchdog
- 弱点：利用ret
- 傻傻的CPU IP寄存器指向的指令
- C3）的完整指令
- +

Recall: Memory Referencing Bug Example

```
typedef struct {
  int a[2];
  double d;
} struct_t;

double fun(int i) {
  volatile struct_t s;
  s.d = 3.14;
  s.a[i] = 1073741824; /* Possibly out of bounds */
  return s.d;
}
```

```
fun(0)  ->   3.1400000000
fun(1)  ->   3.1400000000
fun(2)  ->   3.1399998665
fun(3)  ->   2.0000006104
fun(4)  ->   Segmentation fault
fun(8)  ->   3.1400000000
```

Why?

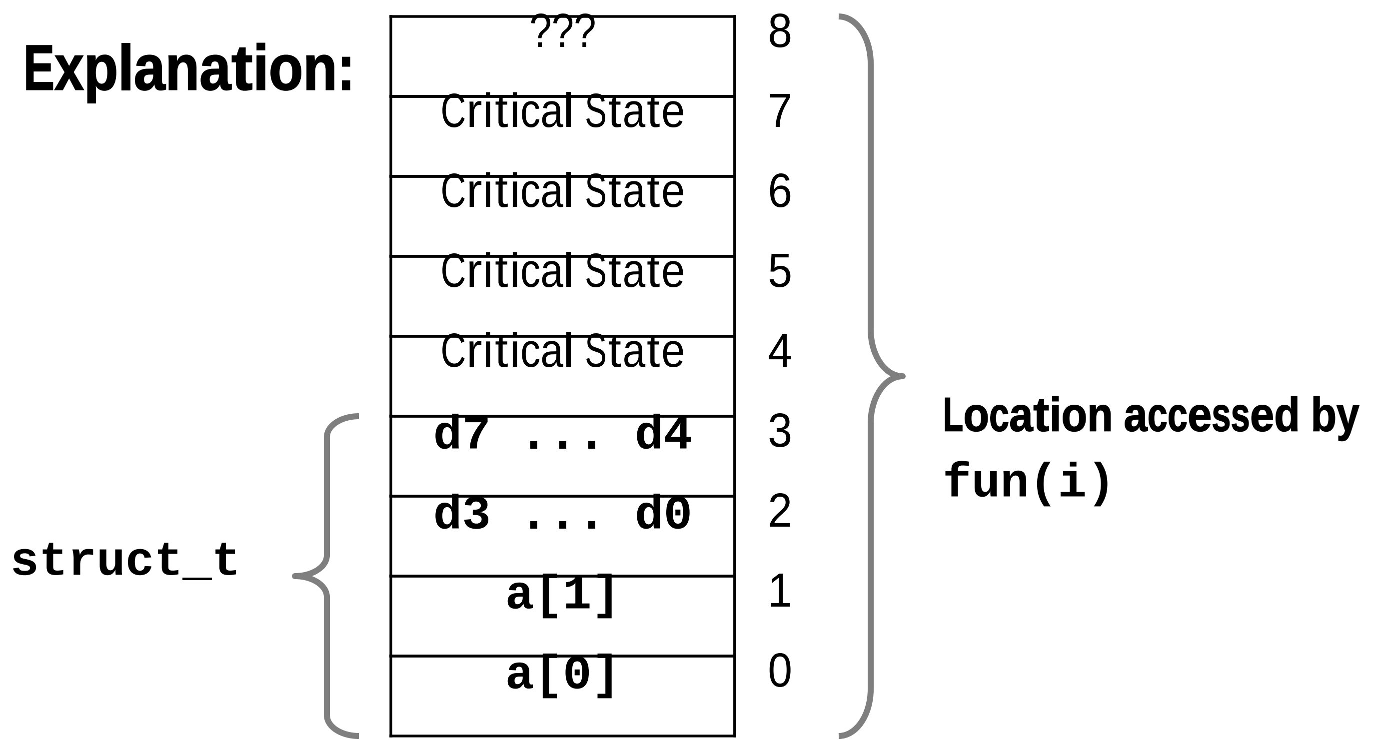
- Result is **system specific**

Memory Referencing Bug Example

```
typedef struct {
  int a[2];
  double d;
} struct_t;
```

```
fun(0)  ->   3.1400000000
fun(1)  ->   3.1400000000
fun(2)  ->   3.1399998665
fun(3)  ->   2.0000006104
fun(4)  ->   Segmentation fault
fun(8)  ->   3.1400000000
```

Explanation:



Such problems are a BIG deal

- **Generally called a "buffer overflow"**
  - when exceeding the memory size allocated for an array
- **Why a big deal?**
  - It's the #1 technical cause of security vulnerabilities
    - #1 overall cause is social engineering / user ignorance
- **Most common form**
  - Unchecked lengths on string inputs
  - Particularly for bounded character arrays on the stack
    - sometimes referred to as stack smashing

String Library Code

- **Implementation of Unix function `gets()`**

```
/* Get string from stdin */
char *gets(char *dest)
{
int c = getchar();
    char *p = dest;
    while (c != EOF && c != '\n') {
        *p++ = c;
        c = getchar();
    }
    *p = '\0';
    return dest;
}
```

  - No way to specify limit on number of characters to read

- **Similar problems with other library functions**
  - **`strcpy`, `strcat`**: Copy strings of arbitrary length
  - **`scanf`, `fscanf`, `sscanf`**, when given **`%s`** conversion specification

Vulnerable Buffer Code

```
/* Echo Line */
void echo()
{
    char buf[4];  /* Way too small! */
    gets(buf);
    puts(buf);
}
```

← **btw, how big is big enough?**

```
void call_echo() {
    echo();
}
```

```
unix>./bufdemo-nsp
Type a string:01234567890123456789012
01234567890123456789012
```

```
unix>./bufdemo-nsp
Type a string:012345678901234567890123
012345678901234567890123
Segmentation Fault
```

Buffer Overflow Disassembly

echo:

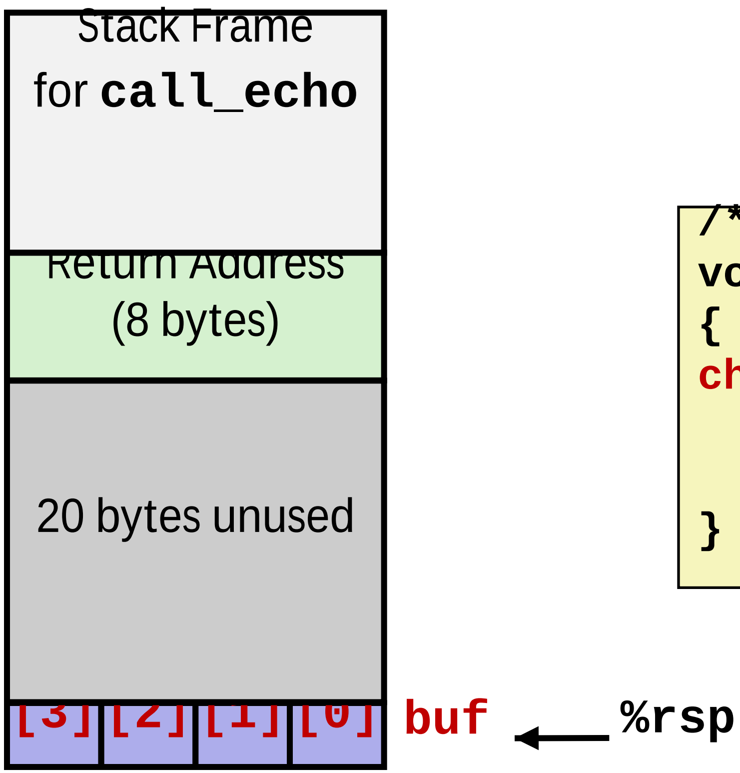
```
 0000000004006cf <echo>:
  4006cf:     48 83 ec 18                      sub    $0x18,%rsp
  4006d3:     48 89 e7                         mov    %rsp,%rdi
  4006d6:     e8 a5 ff ff ff                   callq  400680 <gets>
  4006db:     48 89 e7                         mov    %rsp,%rdi
  4006de:     e8 3d fe ff ff                   callq  400520
<puts@plt>
  4006e3:     48 83 c4 18                      add    $0x18,%rsp
  4006e7:     c3                               retq
```

call_echo:

```
  4006e8:     48 83 ec 08                      sub    $0x8,%rsp
  4006ec:     b8 00 00 00 00                   mov    $0x0,%eax
  4006f1:     e8 d9 ff ff ff                   callq  4006cf
<echo>
  4006f6:     48 83 c4 08                      add    $0x8,%rsp
  4006fa:     c3                               retq
```

Buffer Overflow Stack

Before call to gets

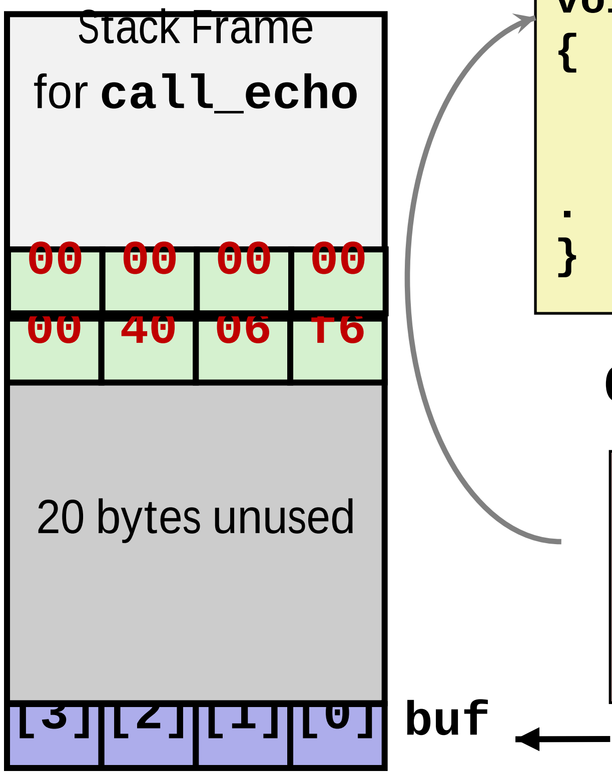



```
/* Echo Line */
void echo()
{
char buf[4];  /* Way too small! */
    gets(buf);
    puts(buf);
}
```

```
echo:
  subq   $0x18, %rsp
  movq   %rsp, %rdi
  call   gets
  . . .
```

Buffer Overflow Stack Example

Before call to gets



%rsp

```
void echo()
{
    char buf[4];
    gets(buf);
. . .
}
```

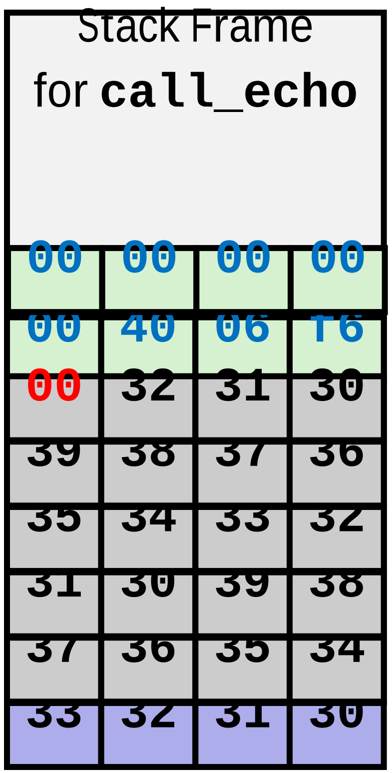
```
echo:
  subq   $24, %rsp
  movq   %rsp, %rdi
  call   gets
  . . .
```

call_echo:

```
 . . .
 4006f1:    callq  4006cf <echo>
 4006f6:    add    $0x8,%rsp
 . . .
```

Buffer Overflow Stack Example #1

After call to gets



buf ← %rsp

```
void echo()
{
    char buf[4];
    gets(buf);
. . .
}
```

```
echo:
  subq   $24, %rsp
  movq   %rsp, %rdi
  call   gets
  . . .
```

call_echo:

```
  . . .
  4006f1:     callq   4006cf <echo>
  4006f6:     add     $0x8,%rsp
  . . .
```

```
unix>./bufdemo-nsp
Type a string:01234567890123456789012
01234567890123456789012
```

"01234567890123456789012\0"

Overflowed buffer, but did not corrupt state

Buffer Overflow Stack Example #2

After call to gets

Stack Frame for `call_echo`			
00	00	00	00
00	40	06	00
33	32	31	30
39	38	37	36
35	34	33	32
31	30	39	38
37	36	35	34
33	32	31	30

buf ← %rsp

```
void echo()
{
    char buf[4];
    gets(buf);
. . .
}
```

```
echo:
  subq   $24, %rsp
  movq   %rsp, %rdi
  call   gets
  . . .
```

call_echo:

```
  . . .
  4006f1:     callq  4006cf <echo>
  4006f6:     add    $0x8,%rsp
  . . .
```

```
unix>./bufdemo-nsp
Type a string:012345678901234567890123
012345678901234567890123
Segmentation fault
```

Program "returned" to 0x0400600, and then crashed.

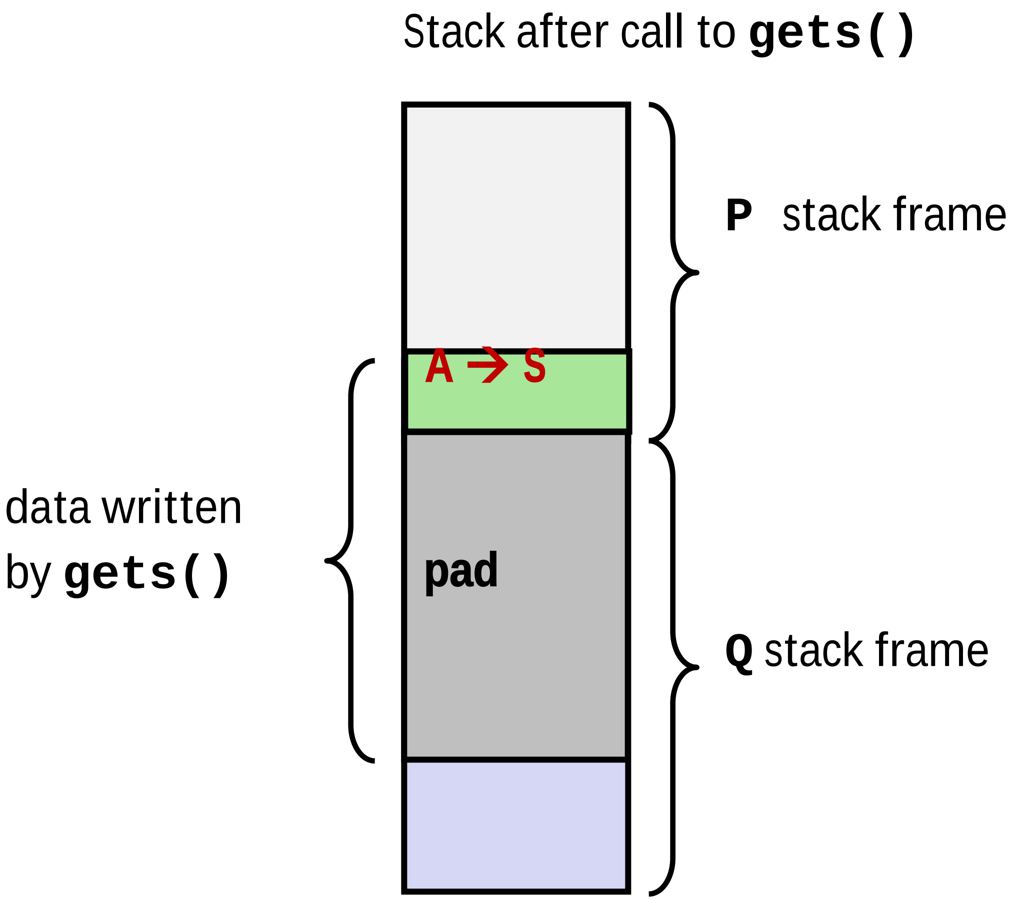
Stack Smashing Attacks

```
void P(){
   Q();
   ...
}
```

return address A

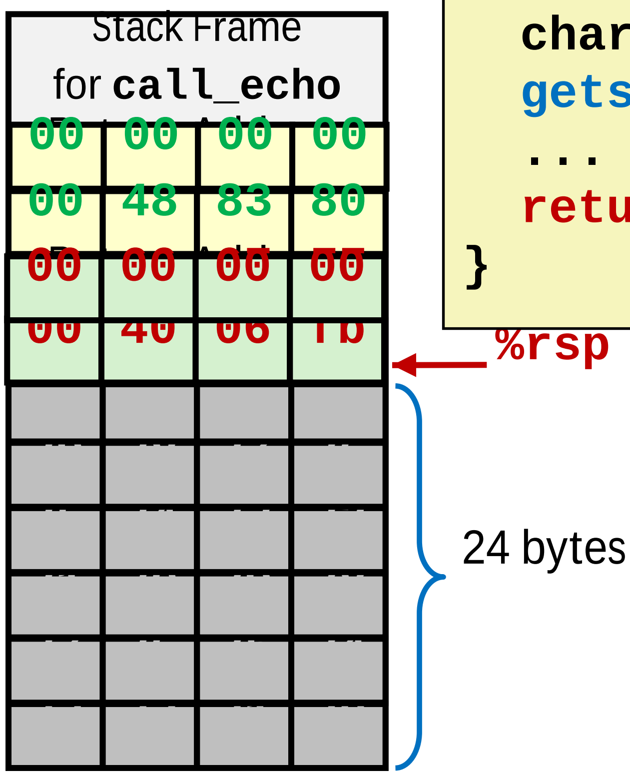
```
int Q() {
   char buf[64];
   gets(buf);
   ...
   return ...;
}
```

```
void S(){
/* Something
   unexpected */
   ...
}
```



- **Overwrite normal return address A with address of some other code S**
- **When Q executes `ret`, will jump to other code**

Crafting Smashing String




```
int echo() {
   char buf[4];
   gets(buf);
   ...
   return ...;
}
```

Target Code

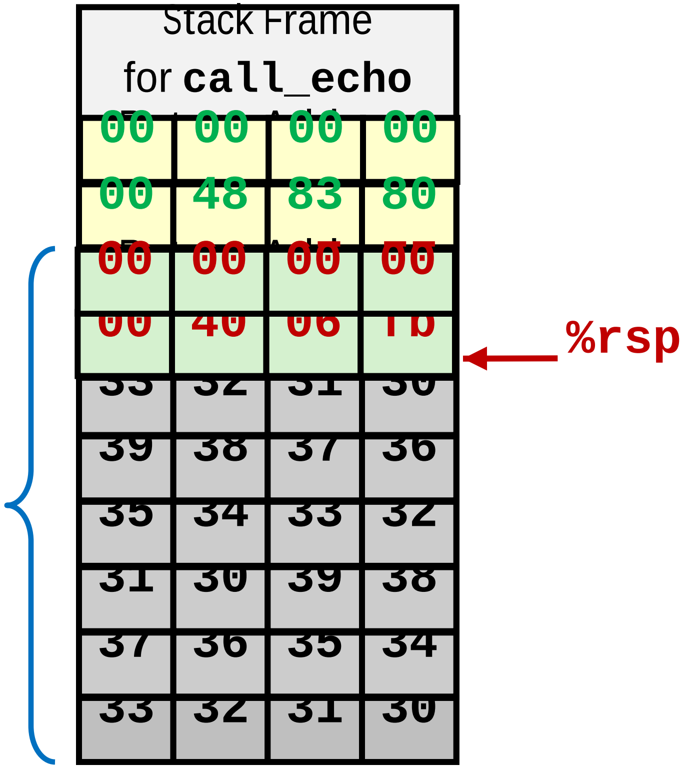
```
void smash() {
   printf("I've been smashed!\n");
   exit(0);
}
```

```
00000000004006fb <smash>:
  4006fb:        48 83 ec 08
```

Attack String (Hex)

```
30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33
fb 06 40 00 00 00 00 00
```

Smashing String Effect



Target Code

```
void smash() {
  printf("I've been smashed!\n");
  exit(0);
}
```

```
00000000004006fb <smash>:
  4006fb:       48 83 ec 08
```

Attack String (Hex)

```
30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33
fb 06 40 00 00 00 00 00
```

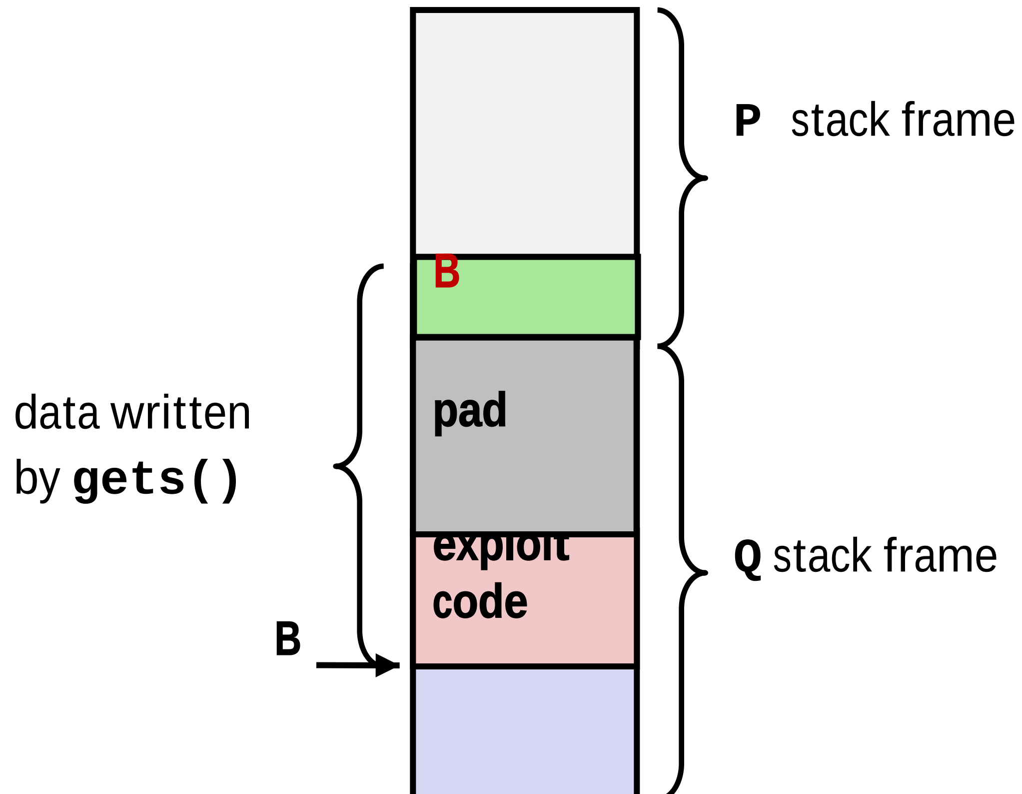
Code Injection Attacks

```
void P(){
   Q();
   ...
}
```

return address A

```
int Q() {
   char buf[64];
   gets(buf);
   ...
   return ...;
}
```

Stack after call to **`gets()`**

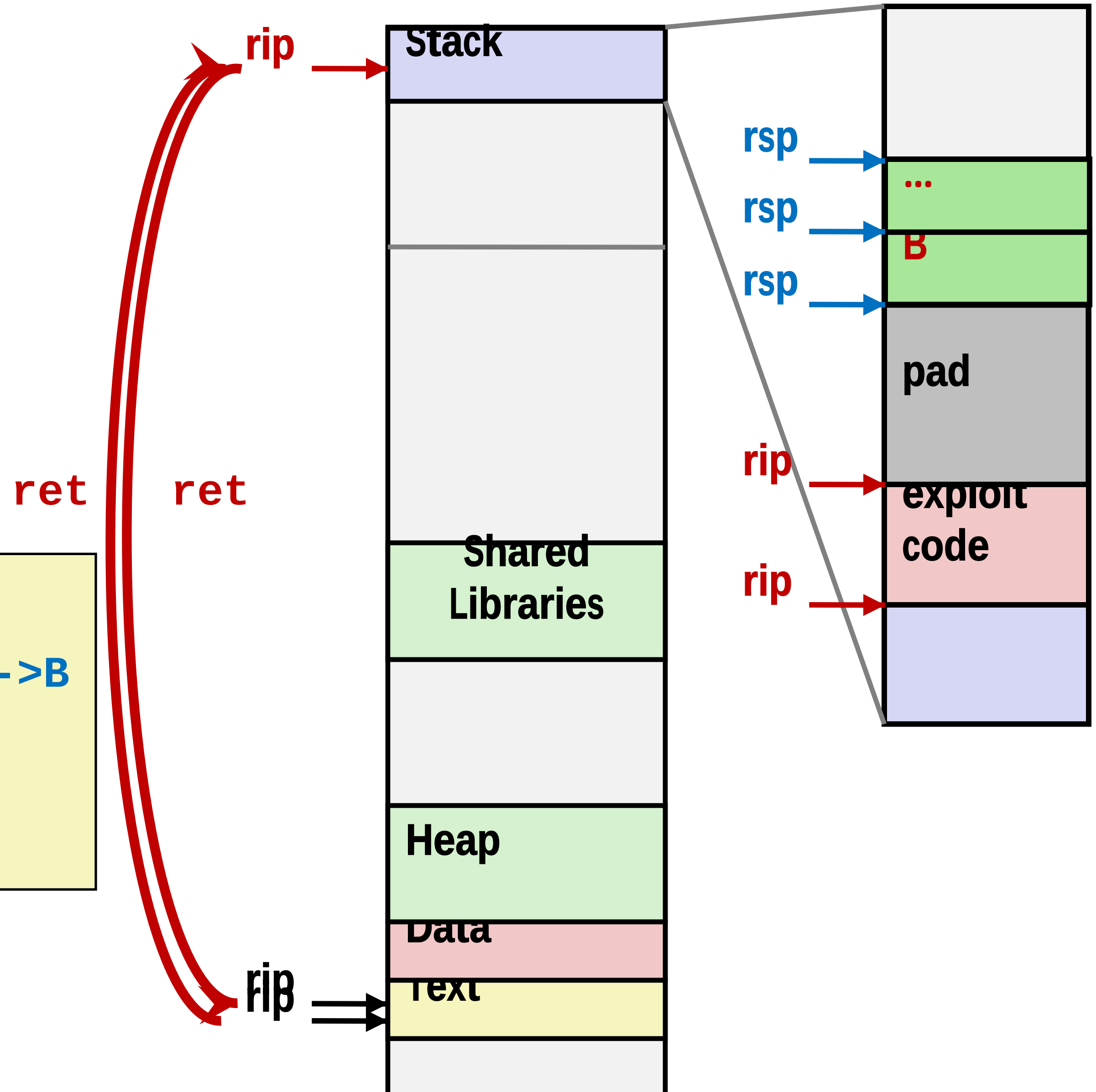


- **Input string contains byte representation of executable code**
- **Overwrite return address A with address of buffer B**
- **When Q executes `ret`, will jump to exploit code**

How Does The Attack Code Execute?

```
void P(){
  Q();
  ...
}
```

```
int Q() {
  char buf[64];
  gets(buf); // A->B
  ...
  return ...;
}
```



What To Do About Buffer Overflow Attacks

- Avoid overflow vulnerabilities
- Employ system-level protections
- Have compiler use "stack canaries"
- Lets talk about each…

1. Avoid Overflow Vulnerabilities in Code (!)

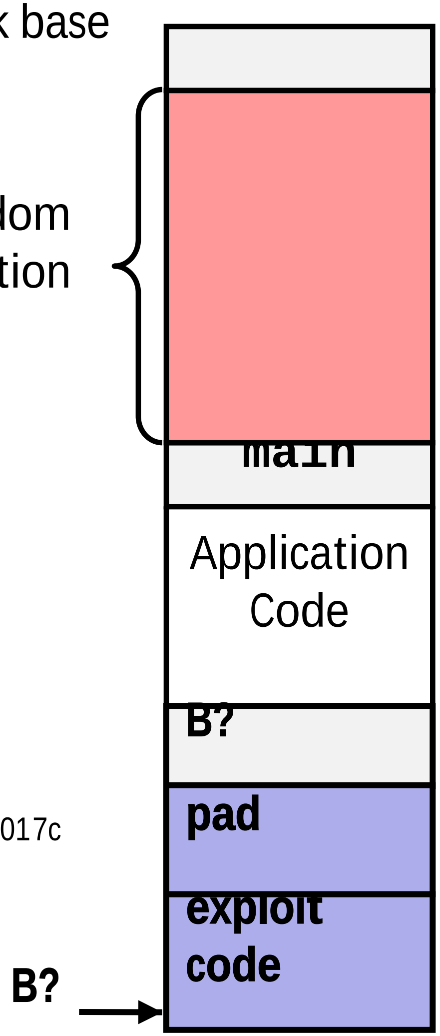
```
/* Echo Line */
void echo()
{
    char buf[4];  /* Way too small! */
    fgets(buf, 4, stdin);
    puts(buf);
}
```

- **For example, use library routines that limit string lengths**
  - **`fgets`** instead of **`gets`**
  - **`strncpy`** instead of **`strcpy`**
  - Don't use **`scanf`** with **`%s`** conversion specification
    - Use **`fgets`** to read the string
    - Or use **`%ns`** where **`n`** is a suitable integer

2. System-Level Protections can help

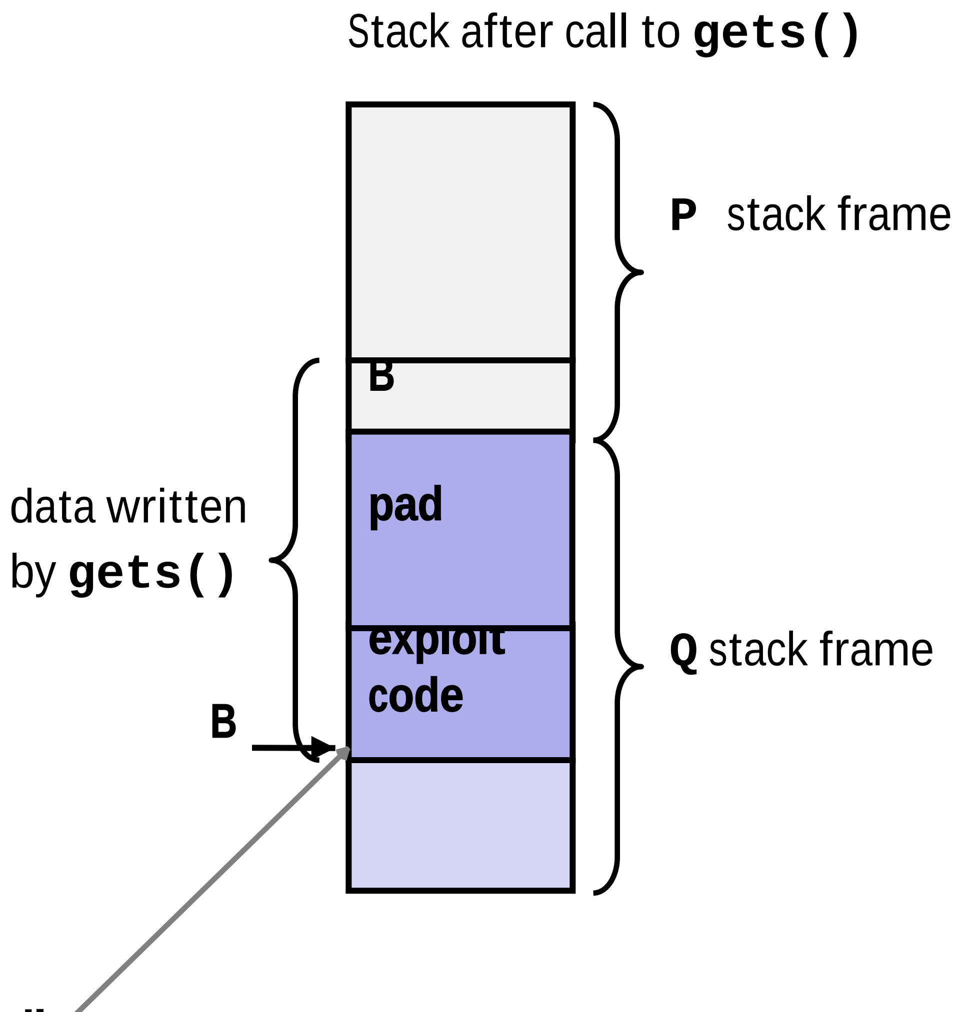
- **Randomized stack offsets**
  - At start of program, allocate random amount of space on stack
  - Shifts stack addresses for entire program
  - Makes it difficult for hacker to predict beginning of inserted code
  - E.g.: 5 executions of memory allocation code
    - Stack repositioned each time program executes

local	0x7ffe4d3be87c	0x7fff75a4f9fc	0x7ffeadb7c80c	0x7ffeaea2fdac	0x7ffcd452017c



2. System-Level Protections can help

- **Nonexecutable code segments**
  - In traditional x86, can mark region of memory as either "read-only" or "writeable"
    - Can execute anything readable
  - x86-64 added explicit "execute" permission
  - Stack marked as non-executable



Any attempt to execute this code will fail

3. Stack Canaries can help

- **Idea**
  - Place special value ("canary") on stack just beyond buffer
  - Check for corruption before exiting function
- **GCC Implementation**
  - `-fstack-protector`
  - Now the default (disabled earlier)

```
unix>./bufdemo-sp
Type a string:0123456
0123456
```

```
unix>./bufdemo-sp
Type a string:01234567
*** stack smashing detected ***
```

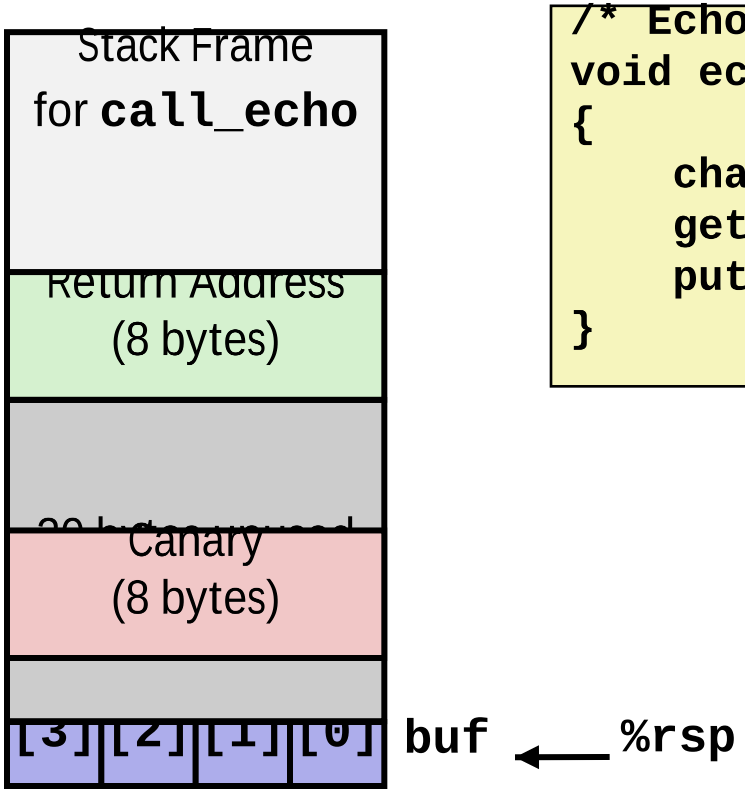
Protected Buffer Disassembly

echo:

```
40072f:     sub    $0x18,%rsp
400733:     mov    %fs:0x28,%rax
40073c:     mov    %rax,0x8(%rsp)
400741:     xor    %eax,%eax
400743:     mov    %rsp,%rdi
400746:     callq  4006e0 <gets>
40074b:     mov    %rsp,%rdi
40074e:     callq  400570 <puts@plt>
400753:     mov    0x8(%rsp),%rax
400758:     xor    %fs:0x28,%rax
400761:     je     400768 <echo+0x39>
400763:     callq  400580 <__stack_chk_fail@plt>
400768:     add    $0x18,%rsp
40076c:     retq
```

Setting Up Canary

Before call to gets

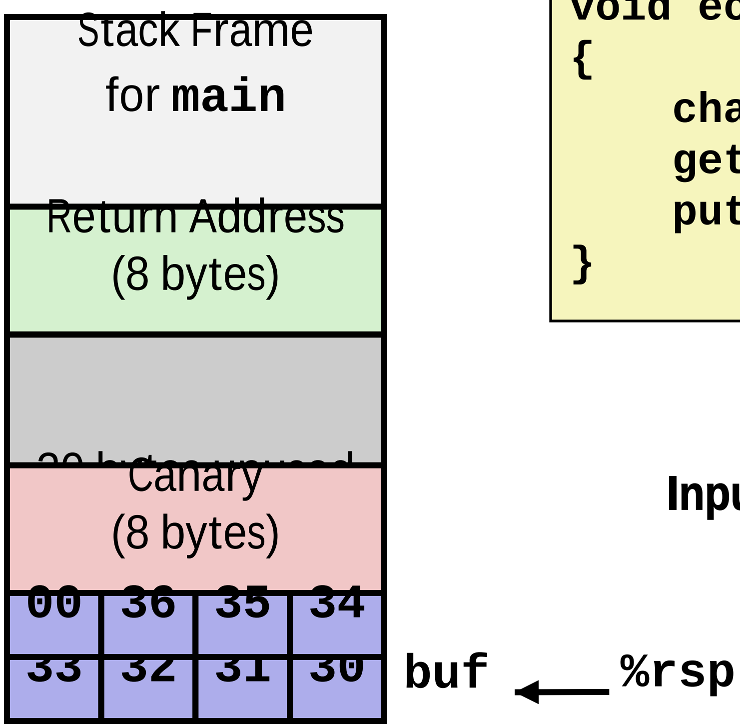



```
/* Echo Line */
void echo()
{
    char buf[4];  /* Way too small! */
    gets(buf);
    puts(buf);
}
```

```
echo:
      . . .
      movq    %fs:40, %rax   # Get canary
      movq    %rax, 8(%rsp)  # Place on stack
      xorl    %eax, %eax     # Erase canary
      . . .
```

Checking Canary

After call to gets




```
/* Echo Line */
void echo()
{
    char buf[4];  /* Way too small! */
    gets(buf);
    puts(buf);
}
```

Input: *0123456*

```
echo:
        . . .
        movq    8(%rsp), %rax     # Retrieve from
stack
        xorq    %fs:40, %rax      # Compare to
canary
        je      .L6               # If same, OK
        call    __stack_chk_fail  # FAIL
```

Return-Oriented Programming Attacks

- **Challenge (for hackers)**
  - Stack randomization makes it hard to predict buffer location
  - Marking stack nonexecutable makes it hard to insert binary code
- **Alternative Strategy**
  - Use existing code
    - E.g., library code from stdlib
  - String together fragments to achieve overall desired outcome
  - *Does not overcome stack canaries*
- **Construct program from *gadgets***
  - Sequence of instructions ending in **`ret`**
    - Encoded by single byte **`0xc3`**
  - Code positions fixed from run to run
  - Code is executable

Gadget Example #1

```
long ab_plus_c
  (long a, long b, long c)
{
   return a*b + c;
}
```

```
00000000004004d0 <ab_plus_c>:
  4004d0:  48 0f af fe  imul %rsi,%rdi
  4004d4:  48 8d 04 17  lea (%rdi,%rdx,1),%rax
  4004d8:  c3           retq
```

rax ← rdi + rdx

Gadget address = `0x4004d4`

- **Use tail end of existing functions**

Gadget Example #2

```
void setval(unsigned *p) {
    *p = 3347663060u;
}
```

Encodes `movq %rax, %rdi`

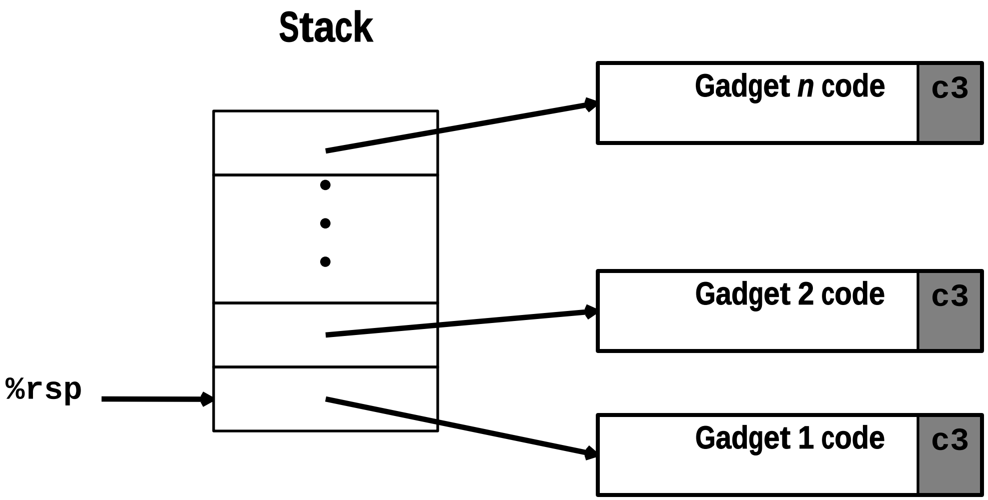
```
<setval>:
  4004d9:  c7 07 d4 48 89 c7  movl  $0xc78948d4,(%rdi)
  4004df:  c3                 retq
```

rdi ← rax

Gadget address = `0x4004dc`

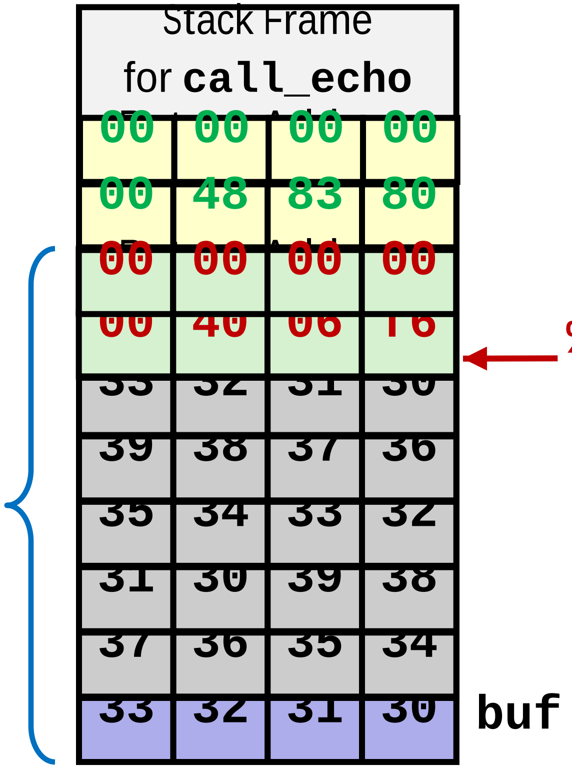
- **Repurpose byte codes**

ROP Execution



- **Trigger with `ret` instruction**
  - Will start executing Gadget 1
- **Final `ret` in each gadget will start next one**

Crafting an ROB Attack String



Gadget

rax ← rdi + rdx

```
0000000000004004d0 <ab_plus_c>:
  4004d0:  48 0f af fe   imul %rsi,%rdi
  4004d4:  48 8d 04 17   lea (%rdi,%rdx,1),%rax
  4004d8:  c3            retq
```

Attack: `int echo()` returns rdi + rdx

```
int echo() {
  char buf[4];
  gets(buf);
  ...
  return ...;
}
```

Attack String (Hex)

```
30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33
d4 04 40 00 00 00 00 00
```

Multiple gadgets will corrupt stack upwards

Exploits Based on Buffer Overflows

- ***Buffer overflow bugs can allow remote machines to execute arbitrary code on victim machines***
- **Distressingly common in real programs**
  - Programmers keep making the same mistakes ☹
  - Recent measures make these attacks much more difficult
- **Examples across the decades**
  - Original “Internet worm” (1988)
  - “IM wars” (1999)
  - Twilight hack on Wii (2000s)
  - … and many, many more
- **You will learn some of the tricks in attacklab**
  - Hopefully to convince you to never leave such holes in your programs!!